Burn ICU Case Study: Marcus Delaney, a Three-Day ICU Stay, Shift by Shift

Critical Care / Advanced Med-Surg | Individual take-home clinical care-plan case study | Open book

Your patient

You are the Burn ICU nurse for Marcus Delaney, a 34-year-old man in room BICU-6 with burns from a house fire. Open the patient in the student EHR (Student View, or the patient file from your instructor). He arrived in the ICU at 0400 on Monday 11/02/26 and the night nurse has cared for him since. You take over at the 0700 shift change, and the chart runs until his transfer to a medical-surgical unit on Thursday 11/05/26 at 0700.

You will work through his stay one 12-hour shift at a time. At the start of each shift, set the simulation clock to the time written at the top of that shift and read the chart up to that moment. You may not need to look past the end of the shift you are working on. Each question tells you where to look in the EHR.

The six shifts

Shift 1: Monday 11/02, 0700 to 1900 (day). Shift 2: Monday 1900 to Tuesday 11/03 0700 (night). Shift 3: Tuesday 0700 to 1900 (day).

Shift 4: Tuesday 1900 to Wednesday 11/04 0700 (night). Shift 5: Wednesday 0700 to 1900 (day). Shift 6: Wednesday 1900 to Thursday 11/05 0700 (night, ends with transfer to Med-Surg).

How this case study works

Each numbered question is a real nursing decision or a piece of reasoning you would do on the unit: reading the flowsheets, labs, orders, MAR and notes, doing the math, and explaining what you would do and why. Write your answer in the large box under each question. Short, specific answers that use Marcus's own numbers earn more points than general textbook answers.

Expect to spend about one hour on each shift, six hours in all. It is a take-home, individual, open-book assignment: use your textbooks, drug guide, notes and the chart.

Scoring

- The whole case study is worth 100 points. A score of 78 or higher is passing.
- Each numbered question shows its point value. Points are given only for what is written in the box.

Writing and turning it in (your pages will be scanned)

- Write in dark pen or very dark pencil, print clearly, and stay inside the box. Write the answer in the box with the same number as the question.
- Write your name and date at the top of every page (the header prints a line for it).
- If you need more room, write "continued on the back" and put the question number on the back.
- Spell out drug names. Show your math: write the formula, the numbers and the units.
- Do not take photos of your work. Your instructor will scan it.
- TURN-IN: put ALL the pages in order and fasten them with a PAPERCLIP. Do NOT staple (the pages must come apart to be scanned). Leave the packet in the instructor's cubby at office G218i.

Honor statement: I completed this work myself, using only my textbooks, notes, drug references and the patient chart.

Signature: _____ Date: _____

PART 1: SHIFT 1: MONDAY 11/02, 0700 TO 1900 (DAY SHIFT)

Nursing process step: Day 1 of the stay | About 60 minutes | 20 points

Set the simulation clock to 0700 on Monday 11/02/26 and move it forward as you work. Marcus arrived in the ICU at 0400; you take report from the night nurse and care for him until you hand off at 1900. You can read everything charted before 0700, and the chart fills in as the clock moves forward.

1.1 Why Is He Here? Your Picture of Marcus at 0700 *(2.5 pts)*

Set the clock to 0700. You are taking report from the night nurse. Read the ED Provider Note, the Burn Surgery and Critical Care H&Ps, the nursing notes from 0425 to 0650 and the Patient Summary. In your own words, explain what happened to Marcus, what injuries he has and what has already been done, then name the THREE most urgent problems you must manage over the next 12 hours.

Where to look in the EHR: Patient Summary; Chart Review > Notes (ED Provider Note, Burn Surgery H&P, Critical Care H&P, Nursing Admission Note)

1.2 The Parkland Formula *(2 pts)*

Find the burn size and his weight in the chart. (a) Calculate his total 24-hour fluid need with the Parkland formula. (b) How much must he receive in the FIRST 8 hours, and by what clock time does that period end? (c) EMS gave 500 mL of lactated Ringer's before the ED. Explain how that changes the hourly rate for the first 8 hours, counting from his ED arrival at 0230. (d) What rate is the second half of the fluid given at? Show your work with the units. Why is the clock started at the time of the injury and not at his arrival?

Where to look in the EHR: Patient Summary (weight); Burn Surgery H&P (TBSA); Orders (lactated Ringer's); EMS Handoff Summary

1.3 Vital Sign Trend *(1.5 pts)*

Copy the vital signs at each time (he has an arterial line, so use the blood pressure shown). Circle every abnormal value. Then, in one or two sentences, describe the trend over the shift and say which value on the flowsheet you do NOT fully trust and why.

Where to look in the EHR: Flowsheets > Vital Signs

Vital sign	Mon 0700	Mon 1100	Mon 1500	Mon 1900
Temperature				
Heart rate				
Blood pressure				
Respiratory rate				
SpO2 (%)				
Pain (CPOT)				

Trend in one sentence:

1.3b Trend (scored with 1.3) *(0 pts)*

Describe the trend over the shift and name the value you do NOT fully trust and why.

1.4 The Pulse Oximeter Says 100% *(1.5 pts)*

Marcus's SpO2 reads 99 to 100% all morning. (a) Explain why this number can be falsely reassuring in this patient. (b) Which lab results show how much carbon monoxide is on his hemoglobin, and what were the values at 0250, 0415 and 0800? (c) What treatment lowers it, and what did the night team change at 0600 (read their notes)? (d) He smokes a pack a day: what does that tell you about the value he is heading toward?

Where to look in the EHR: Lab Results > Blood Gas (Carboxyhemoglobin, PaO2); Orders (ventilator); Flowsheets > Respiratory

1.5 Titrating the Fluids *(1.5 pts)*

Open the lactated Ringer's order, the MAR and the hourly urine output. (a) What is the urine output goal in mL/h and in mL/kg/h? (b) At 0930 the rate was changed: from what to what, and why? (c) Using the graph, write the urine outputs for 1200-1300 and 1300-1400, say what the order tells you to do, and what the new rate becomes at 1400. (d) Name two things you check when you change the rate and one danger of giving too much fluid.

Where to look in the EHR: Orders > lactated Ringer's (burn resuscitation); MAR (rate changes); Flowsheets > Intake/Output

lactated Ringer's 1,000 mL (burn resuscitation, Parkland)
Medication | Active | R. Okafor, MD

STATUS: Active
FREQUENCY: Continuous, titrate hourly
START: 2026-11-02 02:30
END: 2026-11-04 18:00

Instructions
Lactated Ringer's, rate per protocol: start 450 mL/h (first 8 h from injury), then 225 mL/h (hours 8-24 from injury). TITRATE HOURLY to urine output 30-50 mL/h (0.5 mL/kg/h = 41 mL/h). If UOP under 30 mL/h for 2 consecutive hours increase rate 20-25%; if over 50 mL/h for 2 hours decrease rate 20-25%. Notify burn team for UOP under 20 mL/h or rate over 600 mL/h.

Rationale / Clinical Link
Parkland formula 4 mL x 82 kg x 22% TBSA = 7,216 mL in 24 h from injury (01:30). Half in the first 8 h, half over the next 16 h.

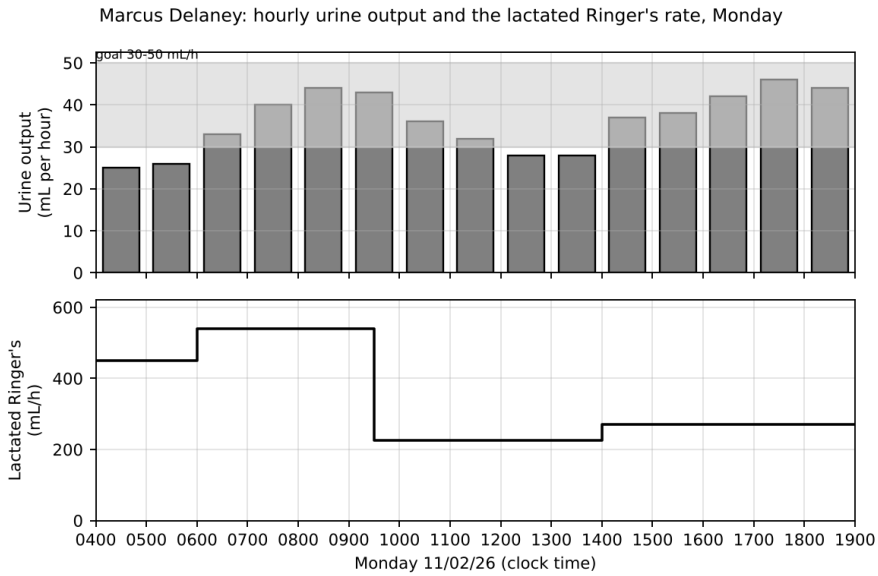
Linked Chart Data
No linked chart data provided.

Nursing Considerations
- Record the rate change and the reason on the flowsheet every hour
- Notify burn team if hourly urine output under 20 mL/h or over 100 mL/h
- Check airway pressures, bladder pressure and right hand pulses with each rate increase

EHR screenshot: Orders, order detail for the lactated Ringer's (click the order to open this panel).

MEDICATION	0700 Mon, 11/2	0800	0900	1000	1100	1200	1300	1400	1500
lactated Ringer's 1,000 mL (burn resuscitation, Parkland) • Dose 450 mL/h 450 mL/h (1,000 mL bag) • IV • Continuous, titrate hourly	0700 Running 540 mL/h		0930 Rate 225 mL/h					1400 Rate 270 mL/h	

EHR screenshot: the MAR row for the lactated Ringer's on Monday (rate changes are shown in the hour they happened).



Graph from the chart: hourly urine output (bars, shaded band is the goal) and the lactated Ringer's rate (line), Monday 0400 to 1900.

1.6 Calling the Burn Surgeon *(1.5 pts)*

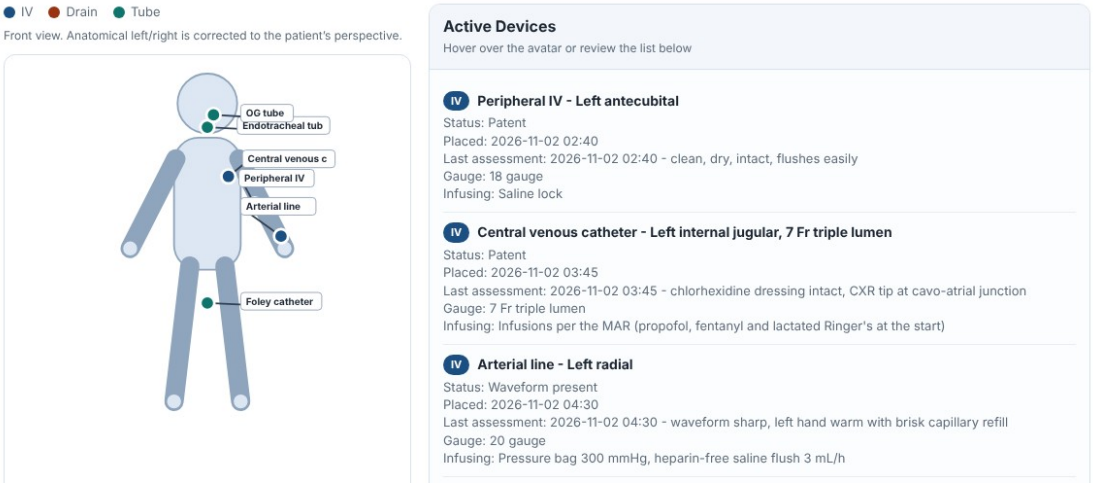
It is 1400 and you have just seen the second low urine output. Write the SBAR you give Dr. Okafor on the phone. Use real numbers from the chart. End with what you are asking him to do and how you will confirm the order.

Where to look in the EHR: Flowsheets (vital signs, I&O, neurovascular); Lab Results (lactate, potassium, CK); Orders

1.7 Lines, Tubes and Airway *(1.5 pts)*

List SIX lines, tubes or devices Marcus has at 0700 (not counting the ventilator). For each, write the site and what you check every shift, and one complication you are watching for.

Where to look in the EHR: Avatar (IVs, drains and tubes); Flowsheets > Lines / Drains / Airways; Nursing Admission Note



EHR screenshot: the Patient Summary page, Avatar and Active Devices list at 0700 (scroll the list to see every device).

Device and site	What you check or do every shift	A complication you watch for

1.8 The Ventilator and Sedation *(1.5 pts)*

Open Flowsheets > Respiratory at 0700. (a) Calculate his ideal body weight (male: 50 kg + 2.3 kg for every inch over 5 feet; 1 inch = 2.54 cm) and the tidal volume at 6 mL/kg. Does this match the set tidal volume? (b) List FOUR actions in the ventilator-associated pneumonia (VAP) bundle that you do this shift. (c) What is the RASS goal on Monday, and why is he sedated more deeply on the first day?

Where to look in the EHR: Flowsheets > Respiratory (vent settings, pressures); Orders (mechanical ventilation; sedation); Orders > Nursing

1.9 Reading the Labs *(1.5 pts)*

Open Lab Results and compare the first draws (about 0240-0250) with the 1200 and 1800 draws. Choose FOUR labs or lab groups that changed or matter. For each, write the values, why this happens in a burn patient, and what you do about it.

Where to look in the EHR: Lab Results (Chemistry, CBC, Blood Gas)

Lab and values (first draw to latest)	Why this happens in Marcus	What you do

1.10 Adding Up Your 12 Hours *(1.5 pts)*

Open Flowsheets > Intake/Output and find the Totals section at the bottom (the screenshot shows where). (a) For the DAY SHIFT 0700 to 1900, what are his total intake, total output and net balance? (b) What was his average urine output in mL/h and in mL/kg/h (82 kg)? (c) Using the Parkland rates, how much lactated Ringer's should he have received BY 1900 counting from the injury (the EMS 500 mL and the ED 675 mL before the ICU count), and how much did he actually receive in all (add up the rates and times on the MAR)? (d) What does that tell you?

Where to look in the EHR: Flowsheets > Intake/Output (Totals at the bottom); MAR (lactated Ringer's); EMS Handoff and ED notes

Blood loss (escharotomy)

TOTALS

Totals	Intake mL	Output mL	Net mL
Night shift 11/01 1900 to 11/02 0700	2349	246	+2103
Day shift 11/02 0700 to 11/02 1900	4720	447	+4273
Night shift 11/02 1900 to 11/03 0700	349	44	+305
Last 24 hours (to now)	**7418**	**737**	**+6681**
Whole stay (matches the Summary page)	**7418**	**737**	**+6681**

EHR screenshot: the Totals section at the bottom of the Intake/Output flowsheet, at 1900 on Monday.

1.11 The 1400 Dressing Change *(1.5 pts)*

Read the Burn Wound Assessment note and the 1520 nursing note. (a) Why was hydromorphone given at 1345 and what tool measures pain in an intubated, sedated patient (name its four parts)? (b) Which topical drugs were applied and where? (c) What do you assess at every dressing change? (d) Name two ways you protect him and yourself from infection.

Where to look in the EHR: Chart Review > Notes (Burn Wound Assessment; Nursing Note 1520); MAR (hydromorphone, silver sulfadiazine, bacitracin)

1.12 Handoff at 1900 *(2 pts)*

It is 1900. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers and name what the night nurse must watch for and do in the first part of the night, including anything coming up in the next few hours.

Where to look in the EHR: Everything you have charted and read this shift; Orders; Lab Results; MAR; Flowsheets

PART 2: SHIFT 2: MONDAY 11/02 1900 TO TUESDAY 11/03 0700 (NIGHT SHIFT)

Nursing process step: Night of day 1 to day 2 | About 60 minutes | 16 points

Set the simulation clock to 1900 on Monday 11/02/26. You take report from the day nurse and care for Marcus until you hand off at 0700. Keep reading forward in the chart as the clock moves; do not look past 0700 for this shift.

2.1 Vital Sign Trend Through the Night *(1.5 pts)*

Set the clock to 1900 on Monday. You take report from the day nurse. Copy the vital signs at each time, circle every abnormal value, then write two sentences below: what the night looked like overall, and what the flowsheet does NOT show about the agitation event that happened around 0245.

Where to look in the EHR: Flowsheets > Vital Signs

Vital sign	Mon 1900	Mon 2300	Tue 0300	Tue 0700
Temperature				
Heart rate				
Blood pressure				
Respiratory rate				
SpO2 (%)				
Pain (CPOT)				

Trend in one sentence:

2.1b Overall night, and what the vitals do not show (scored with 2.1) *(0 pts)*

2.2 Glucose and the Insulin Infusion *(2 pts)*

Open Lab Results (Glucose, point of care and serum), Orders, the MAR and the 0150-0320 nursing notes. (a) List his glucose values from 1800 through 0300 and explain why the insulin infusion was started at 0030 and at what rate. (b) Why does a burn patient have a high glucose even though he has no diabetes? (c) What is the glucose goal and how often do you check it on the infusion? (d) Give TWO safety actions for an insulin infusion, including what you must do if the tube feeding is stopped, and name one lab that you watch closely with insulin.

Where to look in the EHR: Lab Results (Chemistry, Bedside Testing); Orders; MAR (insulin infusion); Chart Review > Notes

2.3 The End of the First 24 Hours *(1.5 pts)*

The Parkland period ends at 0130 (24 hours after the injury). Using the lactated Ringer's rates and times on the MAR (EMS 500 mL, ED 450 mL/h from 0230 to 0400, then the ICU rates), (a) add up the total crystalloid he received from the injury to 0130 and compare it with the Parkland goal. (b) What did the order change to at 0130 and what new fluid starts at 0600? (c) Why do the fluids not simply stop at 24 hours? (d) His weight was 82 kg on arrival and is now 89.6 kg: what does that tell you and what else do you watch for?

Where to look in the EHR: MAR (lactated Ringer's rate changes); Orders; Flowsheets > Cardiac (daily weight) and Intake/Output; Nursing Note 0150

2.4 Alcohol and the Next Three Days *(1 pt)*

His ethanol level was 42 mg/dL at 0240 and he usually drinks 4 to 6 beers on weekends, more than that the night of the fire. (a) When could alcohol withdrawal begin, and when are seizures and delirium tremens most likely? (b) Why may you not see the signs yet? (c) Which two vitamins are ordered and why? (d) Name three signs you would watch for.

Where to look in the EHR: Lab Results (Toxicology); H&P (social history); MAR (thiamine and folic acid)

2.5 The Agitation Event at 0245 *(2 pts)*

Read the 0320 nursing note. Marcus (RASS -3 earlier) becomes restless, thrashes his head, bites the endotracheal tube and the ventilator pressure alarms sound. (a) In order, what do you check and do FIRST, before any medication, and why? (b) The provider orders lorazepam 1 mg IV. Name one concern about giving a benzodiazepine to this patient and one question you could ask the provider. (c) What are the rules for the soft wrist restraint you are using? (d) Besides rechecking RASS, CPOT and vital signs after the dose, what do you document and who do you notify?

Where to look in the EHR: Chart Review > Notes (Nursing Note 0320); Orders (soft wrist restraint, lorazepam); Flowsheets (RASS, CPOT, ventilator); MAR

2.6 Checking the Right Hand and the Abdomen *(1.5 pts)*

Compartment syndrome is a risk after a circumferential burn and a large fluid resuscitation. For each of the six Ps, write what you can or cannot check in Marcus (he is sedated and intubated) and what the chart showed at 2300-0300. Then name the abdominal check ordered every six hours.

Where to look in the EHR: Flowsheets > Musculoskeletal / Mobility (Right Hand Neurovascular check); Nursing Notes; Orders (bladder pressure)

The 6 Ps	Can you check it in Marcus? How?	What the chart showed

2.7 Tube Feeding and Calories *(1.5 pts)*

Open the tube feeding order and the flowsheet. (a) Write the advancing schedule from 1000 Monday and the rate at 0600 on Tuesday. (b) The formula is Osmolite 1.5, which has 1.5 kcal per mL and 62.7 g of protein per liter: how many calories and grams of protein will he receive in 24 hours at 70 mL/h? (c) Give two reasons burn patients are fed early through the gut. (d) What do you check before advancing, and what do you do for a residual over 500 mL?

Where to look in the EHR: Orders (Osmolite 1.5); Flowsheets > GI (Diet / Feeding); Intake/Output; Chart Review (Nutrition Assessment)

2.8 The Midnight and 0600 Labs *(1.5 pts)*

Open Lab Results and compare the 0000 and 0600 draws (and the 0500 albumin and coagulation labs) with the day shift values. Choose FOUR results. Write the values, why they happen in a burn patient on propofol and tube feeding, and what you do or ask for.

Where to look in the EHR: Lab Results (Chemistry, CBC, Coagulation, Blood Gas)

Lab and values	Why this happens in Marcus	What you do or ask for

2.9 Propofol: Math and Safety *(1.5 pts)*

At 1900 on Monday the propofol is running at 35 mcg/kg/min. The bag is propofol 10 mg/mL and his weight is 82 kg. (a) Calculate the rate in mL/h. (b) Propofol gives 1.1 kcal per mL: how many calories does the sedation add in 24 hours? (c) Name two reasons you check the triglyceride level and two signs of propofol infusion syndrome. (d) Name two nursing actions for the bag and tubing.

Where to look in the EHR: MAR (propofol); Orders (propofol); Lab Results (Triglycerides); Flowsheets (RASS)

2.10 Handoff at 0650 *(2 pts)*

It is 0650 on Tuesday. Write your SBAR handoff for J. Morales, RN, the day nurse. Use real numbers, include the events of the night, what is happening in the next few hours (look at the orders and MAR), and anything you would like the day team to follow up.

Where to look in the EHR: Everything you read and charted this night; Orders; MAR; Lab Results; Flowsheets

PART 3: SHIFT 3: TUESDAY 11/03 0700 TO 1900 (DAY SHIFT)

Nursing process step: Day 2: wake-up, wound infection, sepsis | About 60 minutes | 16 points

Set the simulation clock to 0700 on Tuesday 11/03/26. You are J. Morales, RN, and you take report from the night nurse. Keep reading forward in the chart as the clock moves; do not look past 1900 for this shift. A quiet morning is followed by a change in condition in the afternoon.

3.1 The 0800 Sedation Interruption *(1 pt)*

Set the clock to 0700 and read the 0850 nursing note and the 0845 respiratory therapy note. (a) What was done at 0800 and what was the order for the sedatives? (b) How did Marcus respond (RASS, commands, CAM-ICU, his mother)? (c) Name two findings that would make you stop the interruption early and restart the sedation. (d) Why is he NOT having a breathing trial (SBT) today even though FiO2 is 0.40 and PEEP is 5?

Where to look in the EHR: Chart Review > Notes (Nursing Note 0850, Respiratory Therapy Note 0845, ICU Progress Note 0715); Orders (SAT); MAR (propofol)

3.2 Albumin and the Feeds *(1 pt)*

Albumin 5% 500 mL runs from 0600 to 1000. (a) Calculate the pump rate in mL/h. (b) Why is albumin given to Marcus now (use a lab value), and what two things do you assess during the infusion? (c) What did the LR rate change to at 1000? (d) The dietitian wrote about the tube feeding and the 1400 dressing change: what did she recommend about holding the feeding?

Where to look in the EHR: MAR (albumin 5%, lactated Ringer's); Lab Results (Albumin); Nursing Note 1030; Nutrition Follow-Up Note 1045; Flowsheets > Intake/Output

3.3 Noon Electrolytes and the Replacement Orders *(1.5 pts)*

Open the 1200 labs, the orders and the MAR. For each result, write the value, the replacement ordered (drug, dose, route, rate) with the time you gave it, and the safety check or monitoring you do. Then answer below the table: why did the potassium and phosphorus drop when he is on an insulin infusion and tube feeding, and what other 1200 result could the phosphate make worse?

Where to look in the EHR: Lab Results (Chemistry, 1200); Orders; MAR (potassium chloride, magnesium sulfate, sodium phosphate); Nursing Note 1240

Result (1200)	Replacement ordered and time given	Safety check / what you monitor
Potassium		
Magnesium		
Phosphorus		

3.4 The 1400 Dressing Change *(1 pt)*

You and N. Ibarra, RN (burn nurse specialist) change the dressings. Read the 1425 and 1455 notes. (a) List the four NEW findings that told you this is not a normal burn wound. (b) What pain medicine was given, when, and how did CPOT change? (c) What did you do to the edge of the red area at 1430 and why? (d) Which finding at the right chest border is the most worrisome for tissue death?

Where to look in the EHR: Chart Review > Notes (Nursing Note 1425, Wound Care Specialist Note 1455); MAR (hydromorphone); Flowsheets (CPOT)

3.5 Analyze the Graph: Temperature, Heart Rate and MAP *(1.5 pts)*

Study the figure (it is drawn from the flowsheet for 0700 to 1900). (a) At about what time did the change begin and which of the three values moved first? (b) Describe the pattern at 1400, 1700 and 1900 in your own words and say what it means for Marcus. (c) After the norepinephrine starts at 1730 the MAP climbs back up: what does the graph tell you about whether he is getting better? (d) Burn patients often have a fast heart rate and a temperature near 100 F even when they are well: why are the usual SIRS criteria unreliable for him, and name two signs you use instead.

Where to look in the EHR: Flowsheets > Vital Signs (Tuesday 0700-1900); Notes (Critical Care Event Note 1755); the figure below

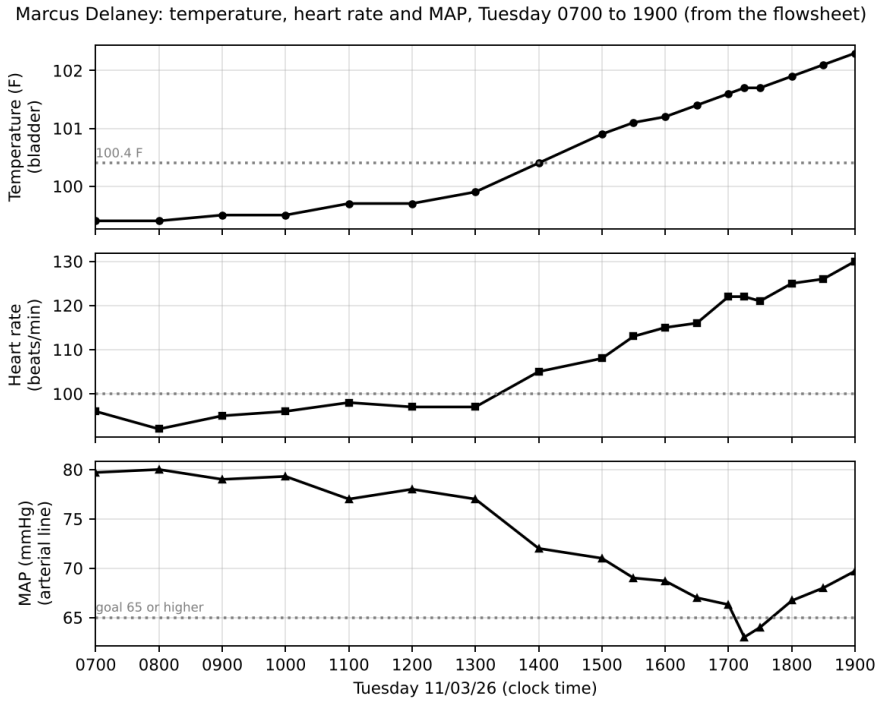


Figure: Marcus Delaney, temperature, heart rate and MAP, Tuesday 0700 to 1900.

3.6 The SBAR Call at 1540 *(1.5 pts)*

It is 1540. You are going to call Dr. Okafor and Dr. Brandt. Write what you say (S, B, A, R) using the chart numbers at 1530 and the trend since 1300. End with what you ask them for: name at least three orders or decisions you request.

Where to look in the EHR: Flowsheets > Vital Signs, Intake/Output, Lab Results; Notes (Nursing Note 1550 Sepsis SBAR)

3.7 The Sepsis Bundle: Cultures and Lactate *(1.5 pts)*

Open the 1655 nursing note and Lab Results. (a) List the four culture sites (five specimens) and the time collected, and which two blood sites. (b) Why must the cultures be drawn BEFORE the antibiotics, and when is it OK to give antibiotics without waiting? (c) Write the lactate values and times from 1555 to 1700 and what they mean. (d) What is the goal for giving antibiotics after recognizing sepsis, and what were Marcus's actual times?

Where to look in the EHR: Chart Review > Notes (Nursing Note 1655, Burn Surgery Event Note 1645); Lab Results (Microbiology, Lactate, Procalcitonin); MAR

3.8 Checking the Antibiotic Orders *(1.5 pts)*

Before you hang the antibiotics you check the orders with pharmacy (A. Chen, PharmD). Marcus weighs 82 kg. (a) The vancomycin load is 20 mg/kg: calculate the dose and compare it with the 1,750 mg on the MAR, then give the pump rate in mg/h for the 2-hour infusion and the order's maximum rate. (b) The piperacillin-tazobactam 4.5 g in 100 mL is an extended infusion: over how long, at what mL/h, and why is it given this way? (c) What does the pip-tazo order say about running it with the vancomycin, and what will you do with the lumens of the left IJ? (d) Name two lab values you follow with this combination and why.

Where to look in the EHR: Orders (piperacillin-tazobactam, vancomycin); MAR; Lab Results (Creatinine); Nursing Note 1655

3.9 Low MAP and Norepinephrine *(1.5 pts)*

At 1715 the BP is 101/44 (MAP 63-64) after the 1700 LR bolus; the second bolus is given at 1745 and norepinephrine 16 mcg/mL is started at 1730 at 0.05 mcg/kg/min. (a) Calculate the pump rate in mL/h for 0.05 mcg/kg/min (82 kg), and the rate at 0.10 mcg/kg/min that you reached at 1830. (b) Where must it run and why (name the complication and one sign)? (c) How and how fast do you titrate, and what is the MAP goal and how is the MAP measured? (d) Why do the providers want to limit more fluid boluses?

Where to look in the EHR: Orders (norepinephrine, LR bolus); MAR; Nursing Note 1755; Critical Care Event Note 1755; Flowsheets > Vital Signs

3.10 Switching the Topical Antimicrobial *(1 pt)*

Silver sulfadiazine was stopped and mafenide acetate 5% soaks were started at 1730. (a) Why did the burn surgeon choose mafenide for this wound? (b) Name the two things you monitor with mafenide and look at the 1700 bicarbonate and the 1845 arterial blood gas: what do they show and why can you not blame the mafenide alone? (c) What do you give before each soak and how often are they done?

Where to look in the EHR: Orders (mafenide, silver sulfadiazine); MAR; Lab Results (Chemistry 1700, Blood Gas 1845, Lactate); Burn Surgery Event Note 1645

3.11 Falling Urine Output *(1 pt)*

The urine output was 45, 41, 36, 28, 24 and 20 mL per hour from 1200 to 1800 (his goal is 0.5 mL/kg/h = 41 mL/h, and Foley care is done). At 1800 the output is 20 mL (0.24 mL/kg/h). Which actions are correct for you now?

Where to look in the EHR: Flowsheets > Intake/Output and Vital Signs; Nursing Note 1240 (potassium plan); MAR; Lab Results (Creatinine); Notes 1755

Select ALL that apply.

- ☐ Report it to Dr. Brandt and Dr. Okafor now (it is below the 30 mL/h call limit) and tie it to the MAP and lactate.
- ☐ Check the Foley for kinks, a full bag tubing loop or sediment, and the bladder pressure (last 12 mmHg).
- ☐ Ask the provider whether to give a third 500 mL LR bolus because he is oliguric, without looking at his lungs or total fluid balance.
- ☐ Hold any further potassium replacement and tell the provider (your 1240 plan was to hold potassium for urine output under 30 mL/h for 2 hours; it has been under 30 since 1500).
- ☐ Ask about the doses of nephrotoxic drugs (vancomycin, pip-tazo) with pharmacy and follow the creatinine.
- ☐ Stop the norepinephrine to see if the urine improves.
- ☐ Keep recording hourly urine output and total I&O; remember he is also getting feeds and fluids, so urine output is the better marker.
- ☐ Give furosemide to make urine because the output is low.

3.12 Handoff at 1850 *(2 pts)*

It is 1850. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers from your shift, say what is changing, and list what the night nurse must do or follow up in the next hours (look at the orders, MAR and labs).

Where to look in the EHR: Everything charted this shift; Orders; MAR; Lab Results; Flowsheets (Vital Signs, Intake/Output); Nursing Note 1850

PART 4: SHIFT 4: TUESDAY 11/03 1900 TO WEDNESDAY 11/04 0700 (NIGHT SHIFT)

Nursing process step: The peak of the burn wound sepsis | About 60 minutes | 16 points

Set the simulation clock to 1900 on Tuesday 11/03/26. You take report from J. Morales, RN, and care for Marcus (burn wound septic shock on norepinephrine) until you hand off at 0700. Keep reading forward in the chart as the clock moves; do not look past 0700 for this shift.

4.1 The Peak of the Sepsis *(1 pt)*

Set the clock to 1930 on Tuesday. Copy the vital signs at each time, circle every abnormal value, then answer below: (a) Calculate the mean arterial pressure at 1930 from the blood pressure and compare it with the MAP on the monitor. (b) Why is a MAP of about 69 NOT reassuring in Marcus at 1930?

Where to look in the EHR: Flowsheets > Vital Signs; Chart Review > Notes (Nursing Note 1950)

Vital sign	Tue 1930	Tue 2300	Wed 0300	Wed 0600
Temperature				
Heart rate				
Blood pressure				
Respiratory rate				
SpO2 (%)				

Trend in one sentence:

4.1b MAP calculation and why 69 is not reassuring (scored with 4.1) *(0 pts)*

4.2 The 1845 Blood Gas *(0.5 pts)*

Open Lab Results (Blood Gas) and the ventilator settings for 1845. (a) Interpret the ABG in one sentence (acid-base disorder and what the body is doing about it) and name the lab drawn at 1900 that explains it. (b) Calculate the PaO2/FiO2 ratio and say what the chest x-ray at 1840 adds.

Where to look in the EHR: Lab Results (Blood Gas, Lactate); Chart Review > Notes (Dr. Brandt 1755, Nursing Note 1850 for the ventilator change); Chart Review > Imaging (Chest X-Ray 1840)

4.3 Norepinephrine: Rate and Safety *(1 pt)*

Open the MAR and the Orders. Norepinephrine 16 mcg/mL was 0.10 mcg/kg/min at 1900, 0.12 at 1930 and 0.14 at 2030. His dosing weight is 82 kg. (a) Calculate the pump rate in mL/h at 0.14 mcg/kg/min. (b) What MAP are you titrating to, and why is the arterial line and a central lumen used? (c) Give two safety checks for a vasopressor infusion.

Where to look in the EHR: MAR (norepinephrine); Orders; Chart Review > Notes (Nursing Note 1755, Dr. Brandt 1755)

4.4 Hydrocortisone, Glucose and the Insulin Infusion *(2 pts)*

Open the MAR, the Orders, Lab Results (point-of-care glucose and potassium) and the pharmacy note of 1930. (a) Hydrocortisone 50 mg IV q6h started at 2030: why would Dr. Brandt order a stress dose, and how many doses did you give this shift? (b) List the glucose values at 1930, 2130 and 0230 and the insulin rate changes between 1900 and 0300, and explain why the glucose rose after the steroid. (c) Why were the insulin rates lowered, and what is the hypoglycemia risk in Marcus? (d) Which electrolyte do you watch with insulin, and what were his values?

Where to look in the EHR: MAR (hydrocortisone, insulin infusion); Lab Results (Glucose, Potassium); Chart Review > Notes (Pharmacy 1930, Dr. Brandt 2345)

4.5 The 2100 Labs and the 2130 Gram Stain *(1 pt)*

Open the CBC at 2100 (compare with 1555 and with admission) and the wound Gram stain at 2130. (a) What do a WBC of 21.8 with 14% bands tell you? (b) Platelets fell from 171 to 142 (and were 288 on admission) while he receives heparin 5,000 units SQ: what percent fall from admission is that, and what do you ask or check before you decide to hold the heparin? (c) What does "Gram-positive cocci in clusters" most likely mean, and do the antibiotics already cover it? (d) What does contact precautions mean for you today?

Where to look in the EHR: Lab Results (CBC with differential, Microbiology); Chart Review > Notes (Result Notification 2140); Orders (Precautions)

4.6 Fever and Cooling *(0.75 pts)*

Open the temperature trend and the MAR (acetaminophen) from 1900 to 0600. (a) Write the temperature at 1930, 2200, 0000, 0300 and 0600 and the times acetaminophen 1 g IV was given. (b) Why record the temperature before you give acetaminophen, and why can you not give all the credit for the falling fever to acetaminophen? (c) Name two cooling measures you used or could use in a burn patient, and one thing you avoid.

Where to look in the EHR: Flowsheets > Vital Signs (Temperature, bladder); MAR (acetaminophen); Chart Review > Notes (Nursing Note 1950); Lab Results (AST, ALT)

4.7 Mafenide Soaks *(0.5 pts)*

The right arm and chest are treated with mafenide 5% soaks every 8 hours (1730, 0130). (a) Why is the pain different with this agent, and how do you score it in Marcus? (b) Mafenide can cause an acid-base problem: which one, and which lab results from this night would you check?

Where to look in the EHR: MAR (mafenide, hydromorphone); Orders; Flowsheets (CPOT); Lab Results (Blood Gas, BMP)

4.8 Sedation and Delirium Risk *(0.5 pts)*

Open the Orders (RASS goal), the flowsheet and the MAR. (a) What is the RASS goal for Tuesday, what was his RASS (charted 2000, 0000 and 0400) on propofol 40 mcg/kg/min and fentanyl 150 mcg/h, and why did Dr. Brandt accept the deeper sedation tonight? (b) Name two delirium risk factors he has and two things you do about them.

Where to look in the EHR: Orders (Sedation and pain scales); Flowsheets (RASS, CAM-ICU, Respiratory); MAR (propofol, fentanyl); Chart Review > Notes (Dr. Brandt 2345)

4.9 Septic Shock That Is Not Responding *(1 pt)*

Choose all that apply.

Where to look in the EHR: Think through the sepsis bundle, MAR (norepinephrine, hydrocortisone) and Orders; this scenario is not in the chart.

HYPOTHETICAL (not in the chart): at 2100 the monitor shows MAP 58 mmHg, HR 134 and urine output 12 mL/h for 2 hours while the norepinephrine is running at 0.14 mcg/kg/min through the left IJ central line. Which nursing actions are appropriate for Marcus?

Select ALL that apply.

- ☐ Notify Dr. Brandt now with SBAR: MAP under 65 on a rising dose of norepinephrine.
- ☐ Check the arterial line (level, zero, compare with a cuff pressure) and confirm the pump rate, concentration and lumen.
- ☐ Turn the norepinephrine down for a few minutes to see whether his pressure recovers on its own.
- ☐ Move the norepinephrine to the left antecubital peripheral IV so the central line can be used for fluid.
- ☐ Expect orders for a second vasopressor (for example vasopressin) and give any ordered fluid in measured boluses, listening to the lungs and watching the ventilator pressures after each one.
- ☐ Give lactated Ringer's wide open until the MAP is above 65, because burn patients always need more fluid.
- ☐ Hold the next hydrocortisone and antibiotic doses until the blood pressure improves.
- ☐ Look for other causes of a falling pressure: bleeding, a tense abdomen (bladder pressure over 15 mmHg), a pneumothorax on the ventilator and the effect of propofol and fentanyl.

☐ Recheck the lactate and glucose, keep hourly urine output and neurovascular checks, and document the times and the notifications.

4.10 The Critical Result at 0240 *(2 pts)*

Open Lab Results (Microbiology) and the nursing note of 0300. At 0240 the microbiology lab calls you. (a) What do you do during the call (what do you say and write down)? (b) Who do you notify, and in what time frame? (c) Write the SBAR you give with the real values. (d) Which line is involved, and what do you assess at the bedside?

Where to look in the EHR: Lab Results (Blood culture 0240); Chart Review > Notes (Critical Result Call 0300, Pharmacy 1930); Flowsheets; Orders (central line care)

4.11 Reading the Pressor Graph *(2 pts)*

Study the graph of MAP, norepinephrine dose and lactate for this shift. (a) What happened to the MAP while the dose was raised from 0.10 to 0.14 and then lowered to zero? (b) How did the lactate change, and what does that mean? (c) What did you need to see before it was safe to wean, and who sets the criteria? (d) After the pressor is off, what do you watch, and what would make you call the provider?

Where to look in the EHR: Flowsheets > Vital Signs (MAP); MAR (norepinephrine); Lab Results (Lactate); Chart Review > Notes (Dr. Brandt 2345, Nursing Note 0450)

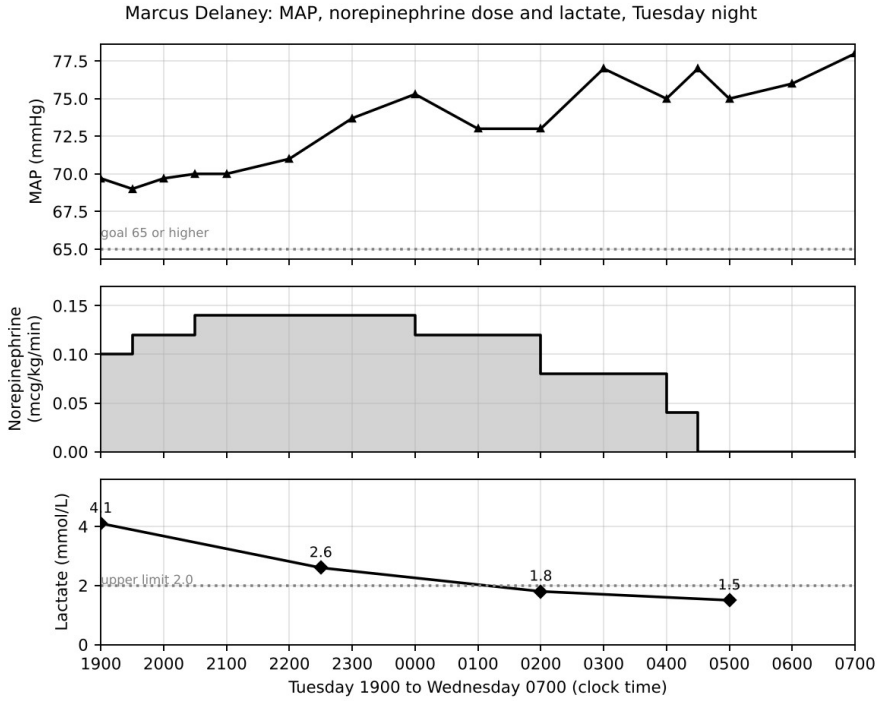


Figure: MAP, norepinephrine dose and lactate, Tuesday 1900 to Wednesday 0700.

4.12 The Kidneys, the Vancomycin Level and Procalcitonin *(1 pt)*

Open Lab Results (Creatinine, Vancomycin level, Procalcitonin) and the pharmacy note. (a) Write the creatinine at 1700, 2300 and 0500 and say what change this is (and whether his urine output agrees). (b) Why does the combination of vancomycin and piperacillin-tazobactam worry the pharmacist? (c) The vancomycin level of 11.6 was drawn at 0415: why is it drawn before the 0430 dose, and what question do you ask? (d) What does procalcitonin 9.6 add?

Where to look in the EHR: Lab Results (Chemistry, Drug levels, Procalcitonin); Chart Review > Notes (Pharmacy 1930, Nursing Notes 0300 and 0450); Intake/Output

4.13 The Night Shift Intake and Output *(0.75 pts)*

Open the Intake/Output flowsheet (Totals) and the albumin order. (a) Write the totals for "Night shift 11/03 1900 to 11/04 0700" and the net balance, and (use the printed shift total, not your own sum of the hourly rows; ignore the later day-shift row) and compare the weight at 0400 (91.2 kg) with the admission weight (82 kg). (b) The output for 0200-0300 is 280 mL: how much is urine and what is the rest? (c) Albumin 5% 500 mL ran 2200 to 0200: what mL/h is that, why was it given, and what do you watch for?

Where to look in the EHR: Flowsheets > Intake/Output (shift totals, hourly rows); Flowsheets (Daily weight, Bowel); MAR (albumin); Orders

4.14 Handoff at 0650 *(2 pts)*

It is 0650 on Wednesday. Write your SBAR handoff for J. Morales, RN, the day nurse. Use real numbers, the events of the night, what is due in the next hours (look at the MAR and Orders), and anything you want the day team to follow up.

Where to look in the EHR: Everything you read and charted this night; Orders; MAR; Lab Results; Flowsheets

PART 5: SHIFT 5: WEDNESDAY 11/04 0700 TO 1900 (DAY SHIFT)

Nursing process step: Recovery and extubation day | About 60 minutes | 16 points

Set the simulation clock to 0700 on Wednesday 11/04/26. You are J. Morales, RN, the day nurse: Marcus is recovering from the burn-wound sepsis, off norepinephrine since 0430, and is due for the sedation break and breathing trial. Keep reading forward in the chart as the clock moves; do not look past 1900 for this shift.

5.1 The 0800 Sedation Break and the 0930 Breathing Trial *(1.5 pts)*

Read the 0820 nursing note, the 0955 respiratory therapy note and the SBT order. (a) What does a PASSED daily sedation interruption (SAT) look like, and what did Marcus show (RASS, CAM-ICU, pain, vital signs)? Which sedative was turned off and which continued? What would a FAILED SAT look like and what do you do? (b) Describe the spontaneous breathing trial (settings and length), then CALCULATE the RSBI from the respiratory rate and the exhaled tidal volume in the RT note and compare it with the pass value. (c) What did the cuff-leak test show and why is it done in a patient with an inhalation injury?

Where to look in the EHR: Chart Review > Notes (0820 Nursing Note, 0955 RT Note); Orders (SBT, SAT); Flowsheets (RASS, CPOT, ventilator)

5.2 Reading the Breathing Graph Before and After Extubation *(1.5 pts)*

The graph shows respiratory rate, SpO2 and heart rate from the flowsheet, Wednesday 0400 to 1200 (the dotted line is 30 breaths/min). He was extubated at 1045. (a) Looking only at 0600 to 1000, what does the graph show about his readiness to come off the ventilator? Use numbers. (b) What changes on the graph after 1045, give the values, and explain why this is or is not worrying. (c) Which graph changes would make you call the provider and RT, and what would you do at the bedside while waiting?

Where to look in the EHR: Flowsheets > Vital Signs (RR, SpO2, HR); Chart Review > Extubation Procedure Note and 1250 Nursing Note (ABG 1200)

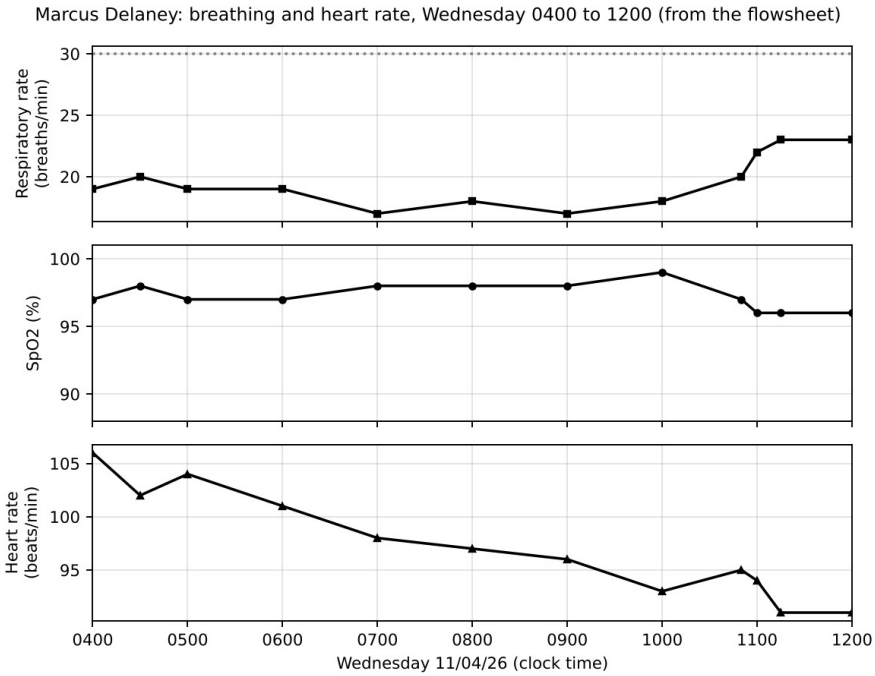


Figure: Respiratory rate, SpO2 and heart rate, Wednesday 11/04/26, 0400 to 1200 (from the flowsheet). Dotted line = 30 breaths/min.

5.3 Right After the Tube Comes Out (1045) *(1.25 pts)*

Open the 1100 Extubation Procedure Note and your 1130 nursing note. (a) List FOUR things you assess in the first hour after extubation (and how often) and what you found at 1050-1115. Why is he at higher risk for airway swelling than another patient? (b) If STRIDOR develops, what do you do and what does the provider plan say to give? (c) Why must the airway watch continue even though he looks well at 1100?

Where to look in the EHR: Chart Review > Notes (1100 Extubation Procedure Note, 1130 Nursing Note, 0955 RT Note); Orders; MAR

5.4 Dexmedetomidine After Extubation *(0.75 pts)*

Dexmedetomidine was started at 1100. Open the order. (a) Calculate the pump rate in mL/h (weight 82 kg, bag 4 mcg/mL, dose 0.4 mcg/kg/h). (b) When do you hold it and notify the provider according to the ORDER? (c) The surgeon wrote "avoid benzodiazepines": give two reasons that a drug like dexmedetomidine suits this patient better than lorazepam.

Where to look in the EHR: Orders and MAR (dexmedetomidine); Chart Review > 1100 and 1130 Notes; Flowsheets (HR, BP, RASS)

5.5 NPO, the OG Tube, Breathing Exercises and Nausea *(0.75 pts)*

After extubation he was NPO, the OG tube was removed at 1115, and at about 1215 he felt nauseated; ondansetron 4 mg IV was given at 1230. (a) Why NPO until the swallow screen, and why was the OG tube taken out now? (b) What do you do for incentive spirometry and oral care? (c) For the nausea: what do you do first, what is the safe-practice trap, and what do you check about the drug?

Where to look in the EHR: Orders (Diet, Aspiration precautions); MAR (ondansetron); Chart Review > 1130 and 1250 Nursing Notes, 1230 Nutrition Note

5.6 The Insulin Infusion Ends at Noon: Reading the Correction Scale *(1 pt)*

Open the glucose results, the insulin lispro correction-scale order and the MAR. (a) His glucose values from 0930 to 1800 and why the infusion was stopped at 1200. (b) Using the ORDER, how many units of lispro would you give for glucose 149, 162, 238, 276 and 362, and when do you call the provider? (c) Name one safety concern in the hours around the change (feeds stopped, hydrocortisone dropped to 25 mg) and how often glucose is checked now.

Where to look in the EHR: Lab Results (Glucose, point of care and serum); Orders (insulin lispro correction scale); MAR; Notes (1230 Nutrition note)

5.7 The 1330 Family Meeting *(0.5 pts)*

Read the Case Management family meeting note (the meeting was at 1330). (a) Give two things Dr. Okafor told the family about the next days. (b) The patient was not in the room: what happened afterwards, and give one thing you as the bedside nurse can do to support Marcus and his family after the meeting.

Where to look in the EHR: Chart Review > Notes (Case Management Family Meeting Note, 1430)

5.8 The 1400 Dressing Change with Mafenide *(1 pt)*

Read the 1430 nursing note and the MAR. (a) When and what did you give before the dressing change, and why that timing? He still reported 6/10: what else do you do? (b) Why is mafenide used for this wound, and what two problems do you watch for with mafenide (find the numbers that show whether the second one is happening)? (c) Name four things you assess and document about the wound at this change, using what the note found (and why the marker line matters). (d) What precautions apply?

Where to look in the EHR: MAR (hydromorphone 1345, mafenide, acetaminophen); Orders (burn wound care); Lab Results (CO2/bicarbonate, ABG 1200); Chart Review > 1430 Nursing Note

5.9 Reading the Culture Results in the Lab Section *(1.25 pts)*

The picture shows the Microbiology rows of Lab Results at 1630 Wednesday (cultures were drawn Tuesday at 1550). (a) For the wound culture, the blood cultures (4 bottles), the tracheal aspirate, the urine culture and the repeat blood cultures, write what each shows and the time column it appears under. Is any result still not final? (b) The rows for blood culture set 1, set 2 and the wound Gram stain look empty: why, and where would you scroll to find their results? (c) The lab called at 1500: what does "MSSA in 2 of 4 bottles" tell you (true infection or contaminant?) and what does it NOT yet tell you? (d) What does "no growth to date" on the repeat blood cultures mean at 1630?

Where to look in the EHR: Lab Results > Microbiology (newest columns at the right; scroll left for the 11/03 1550 and 11/04 0240 entries); Chart Review > 1540 Pharmacy Note

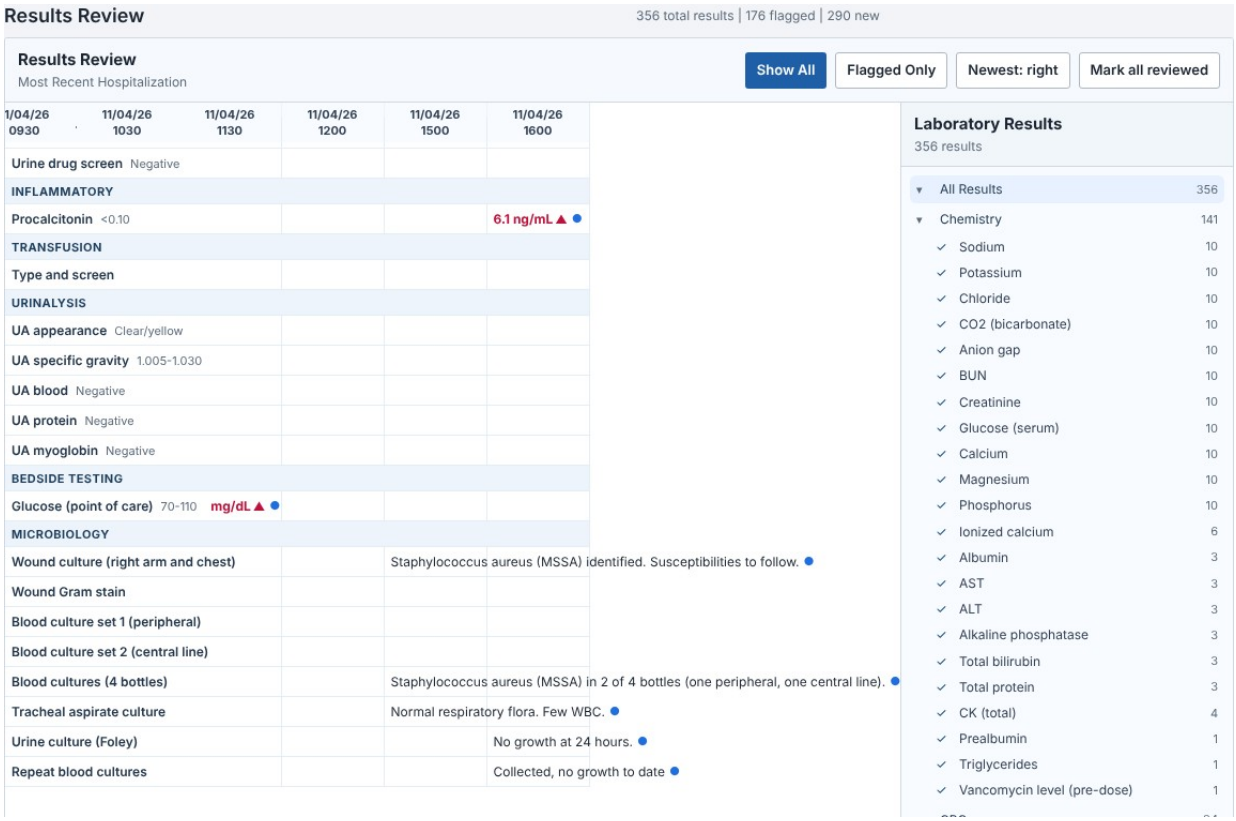


Figure: Lab Results review, Microbiology and Bedside Testing rows, 11/04/26 at 1630 (columns 0930 to 1600).

5.10 MSSA: Narrowing the Antibiotics, Repeat Cultures and the Central Line *(1.5 pts)*

Read the 1145 ID consult, the 1540 pharmacy note, the Orders and the 1645 nursing note. (a) Vancomycin was stopped at 1530: why, and which antibiotic is better for MSSA? (b) Why does narrowing antibiotics matter, and why is pip-tazo still running? (c) What did ID ask for about repeat blood cultures: how many, from where, when (you drew them at 1600 before the 1630 pip-tazo), and for how long will he need antibiotics? (d) What is the decision about the left IJ central line and what would change it? What do you do for the line meanwhile?

Where to look in the EHR: Chart Review > Notes (1145 ID Consultation, 1540 Pharmacy Stewardship Note, 1645 Nursing Note); Orders (vancomycin, cefazolin, pip-tazo, repeat blood cultures); MAR

5.11 First Time Out of Bed with PT and OT *(0.75 pts)*

Read the 1520 PT note, the 1535 OT note and the OOB order. (a) What do you watch and measure during the first dangle, stand and chair transfer, and when do you stop? (b) Give three safe-handling points for the lines and the burned right arm. (c) What are the splint instructions from OT?

Where to look in the EHR: Orders (Activity); Chart Review > PT Note 1520 and OT Note 1535; Flowsheets (HR, SpO2)

5.12 The Swallow Screen, Diet and the New Pain Pills *(0.75 pts)*

At 1600 he passed the bedside swallow screen and the soft diet started at 1800. (a) What did the screen include and what are your aspiration precautions? When do you stop the meal? (b) You give oxycodone 5 mg at 1645: why 5 mg, and what do you check first (look at the order and the other sedating drugs)? (c) Gabapentin 300 mg three times daily was also ordered: why, and what do you watch for?

Where to look in the EHR: Chart Review > 1610 SLP Note, 1645 Nursing Note; Orders (Diet, oxycodone, gabapentin); MAR

5.13 Wednesday Labs: Is the Infection Getting Better? *(0.75 pts)*

Open Lab Results and look at the creatinine, WBC (with bands) and procalcitonin from Tuesday afternoon through the 1700 draw on Wednesday. Write the values for each and what each trend tells you about his kidneys and the infection.

Where to look in the EHR: Lab Results (Chemistry: Creatinine, BUN; CBC; Inflammatory: Procalcitonin)

5.14 Intake and Output: Where Did the Fluid Go? *(0.75 pts)*

Open the Intake/Output flowsheet. The Day shift 11/04 0700 to 1900 total is intake 1,823 mL, output 836 mL, net +987 mL. (a) Calculate his average urine output in mL/h and mL/kg/h (82 kg) and compare it with the Parkland goal of 0.5 mL/kg/h. (b) Why is the urine output high now, and why is the net still positive although the fluids were cut (LR 75 -> 50 mL/h at 1200, stopped at 1800)? (c) What would you watch or ask the team about next?

Where to look in the EHR: Intake/Output flowsheet (Totals and hourly entries); Orders; Flowsheets > Cardiac; MAR (LR rate changes)

5.15 Handoff at 1850 *(2 pts)*

It is 1850 on Wednesday. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers from the chart, include what happened today, what the night nurse must do or watch in the next hours (look at the MAR and orders), and the things still pending.

Where to look in the EHR: Everything you read and charted this shift; Orders; MAR; Lab Results; Flowsheets; Intake/Output

PART 6: SHIFT 6: WEDNESDAY 11/04 1900 TO THURSDAY 11/05 0700 (NIGHT SHIFT)

Nursing process step: Recovery and transfer out of the ICU | About 60 minutes | 16 points

Set the simulation clock to 1900 on Wednesday 11/04/26. You take report from J. Morales, RN, and care for Marcus until you hand him off to the Med-Surg nurse at 0650 on Thursday 11/05. Keep reading forward in the chart as the clock moves; do not look past 0700 for this shift.

6.1 Vital Signs Through the Last Night *(0.75 pts)*

Set the clock to 1900 on Wednesday 11/04. Copy the vital signs at each time and circle every abnormal value. Then write two sentences below: (1) what changed in his breathing support and pain across the night, and (2) one thing the vital signs cannot tell you about the infection.

Where to look in the EHR: Flowsheets > Vital Signs

Vital sign	Wed 1900	Wed 2300	Thu 0300	Thu 0500
Temperature				
Heart rate				
Blood pressure				
Respiratory rate				
SpO2 (%)				
Pain (0-10)				

Trend in one sentence:

6.1b Two sentences for item 6.1 *(0.25 pts)*

(1) What changed in his breathing support and pain across the night? (2) One thing the vital signs cannot tell you about the infection.

Where to look in the EHR: Flowsheets > Vital Signs

6.2 Soft Diet and the Correction Scale *(0.75 pts)*

Marcus eats a soft diet with shakes and the insulin infusion is off. (a) Read the dinner intake in the 2000 flowsheet and the glucose values at 1800, 0000 and 0600; using the lispro correction scale order, how many units were due for each? (b) The last hydrocortisone dose was at 0230: how will that change his glucose, and why is the risk of low glucose still small on this scale? (c) He eats only about half of each tray: name two actions for his nutrition before Friday's surgery.

Where to look in the EHR: Flowsheets > GI (Diet / Feeding); Lab Results (Glucose, point of care); Orders (insulin lispro correction scale, soft diet); Intake/Output

6.3 Pain and Sleep Now That He Is Awake *(1.25 pts)*

Open the 2230 and 0315 nursing notes, the pain flowsheet and the MAR. (a) Which pain scale do you use now, and why is the CPOT no longer the right tool? Give his scores at rest and with turning or the soaks. (b) List the doses of oxycodone, gabapentin and acetaminophen he received overnight with times. At 0130 mafenide soaks hurt (5/10) and the oxycodone was given at 0215: what would you do differently for the next soak at 0930? (c) At 0615 you are about to give the acetaminophen tablet: what must you check on the MAR and the order before giving it, and what is his daily limit? (d) Name two ways you protected his sleep and one safety check you make after giving oxycodone with gabapentin.

Where to look in the EHR: Chart Review > Notes (Nursing Note 2230, 0315); Flowsheets > Pain; MAR; Orders (acetaminophen, oxycodone, gabapentin)

6.4 Delirium and Alcohol Withdrawal Watch *(1.25 pts)*

His drinking history and an ethanol level of 42 mg/dL at 0240 on Monday were charted. Sedation is off, so days 2 to 4 after the last drink are the highest-risk window. (a) Name the four features of the CAM-ICU and say what makes it positive; what are his CAM-ICU and RASS results on this shift? (b) List four signs of alcohol withdrawal or delirium tremens you would look for and which vital signs would change first. (c) Three non-drug actions that prevent delirium tonight. (d) If he became agitated and tremulous at 0400, what would you do first and what would you ask the provider (look at what he was told to avoid)?

Where to look in the EHR: Flowsheets > Neuro (CAM-ICU, RASS); Chart Review > H&P (social history) and Notes; Lab Results (Toxicology); MAR (thiamine and folic acid); Pharmacy Transfer Medication Reconciliation

6.5 Pip-Tazo to Cefazolin *(1.25 pts)*

The last pip-tazo dose is given at 0030 and cefazolin 2 g IV is given at 0610. (a) Compare the two drugs for this patient: what did the blood and wound cultures grow, which drug is the best choice for it, and give TWO reasons for de-escalating from pip-tazo (look at the pharmacy and ID notes and the allergy list). (b) Pip-tazo is an extended infusion: how long does the 0030 dose run? How long is the cefazolin infusion on the order, and what rate in mL/h gives 2 g in 100 mL over that time? (c) Write the next two cefazolin times, and name two things you check before and after the first dose.

Where to look in the EHR: Orders (cefazolin, piperacillin-tazobactam); MAR; Lab Results (Microbiology); Chart Review > Notes (Pharmacy Stewardship Note, Burn Surgery Note); Allergies

6.6 Three Treatments That Stopped Overnight *(1.25 pts)*

Three supports were stopped or weaned during the night. For each one, write when and how it stopped, what the chart showed afterward (look at the 0300 and 0500 vital signs and the notes), and what you watch for next.

Where to look in the EHR: MAR (hydrocortisone, dexmedetomidine); Orders; Flowsheets > Vital Signs and Respiratory; Notes 0315 and 0520

Treatment	When it stopped / how it was tapered	What the chart showed afterward and what you watch for

6.7 The Three-Day Trend Graph *(2 pts)*

Study the graph of temperature, WBC, procalcitonin and creatinine for the whole stay (from the chart). Use the 0430 values on Thursday (WBC 11.2, creatinine 1.0, procalcitonin 2.1). (a) For each of the four lines, state the peak value and when it happened, and the latest value. (b) What does the graph show about the infection and about his kidneys? (c) At what time did he turn the corner, and why do the lab lines peak AFTER the temperature does? (d) What must the Med-Surg team keep watching? Give three things from the graph and the chart.

Where to look in the EHR: Figure below; Lab Results (WBC, Procalcitonin, Creatinine); Flowsheets > Vital Signs (Temperature); Notes (ICU Transfer Summary)

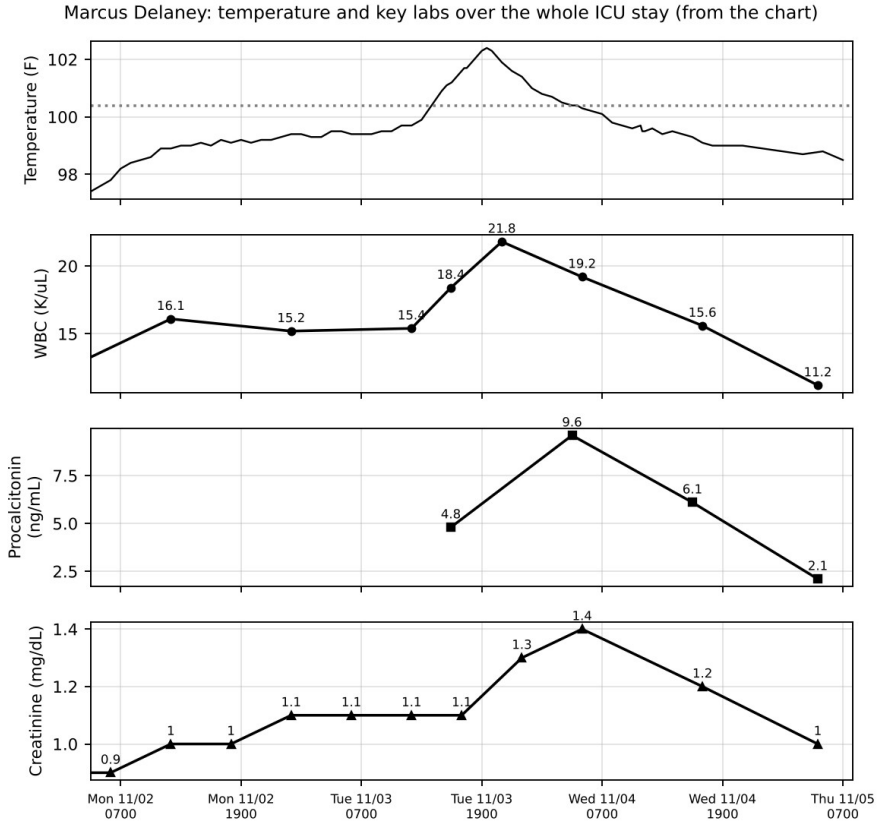


Figure: Temperature, WBC, procalcitonin and creatinine, Monday 0700 to Thursday 0700 (dotted line = 100.4 F).

6.8 The Arterial Line and the Foley Come Out at 0500 *(1.25 pts)*

Read the 0520 nursing note. (a) Describe how you remove the left radial arterial line, how long you hold pressure (and why that could be longer in some patients), and what you check in the left hand and the site afterward and when. Which arm is used for the blood pressure now? (b) The Foley comes out at 0500: what is the target time for the first void, what do you do if he does not void, and what is his urine output goal in mL/h (0.5 mL/kg/h at 82 kg)? (c) Give two CAUTI prevention measures for the catheter days and one reason this one was removed now.

Where to look in the EHR: Chart Review > Notes (Nursing Note 0520); Orders (Foley care and necessity review); Devices; Lab Results (platelets, INR)

6.9 The Central Line on Line Day 4 *(0.75 pts)*

The left IJ central line stays after the arterial line, the PIV and the Foley are gone. (a) Count the line day (look at the Devices page) and explain why ID allowed it to stay even though he had bacteremia. (b) Give three CLABSI-prevention actions you do on this line every shift. (c) What is the one question you would ask the provider to get the line out sooner?

Where to look in the EHR: Flowsheets > Devices and Lines; Orders (Central line care); Notes (Nursing Note 0520, ICU Transfer Summary)

6.10 Is He Ready to Transfer? *(1.25 pts)*

The transfer to Med-Surg 2 East is planned for 0700. For each criterion, write the chart value or time that proves it is met (with the actual number of hours where asked), and in the last row state your decision (ready or not ready) and one concern you would still hand off.

Where to look in the EHR: Flowsheets (Vital Signs, I&O); MAR (norepinephrine); Devices (ETT); Notes (Burn Surgery Transfer Note, Case Management Transfer Note)

Criterion	Evidence from the chart (values and times)

6.11 Fall Risk and Mobility After the Lines Are Out *(0.75 pts)*

The arterial line and Foley are out, he is on gabapentin and oxycodone and his right arm is splinted. Which actions are correct for the rest of this shift and the handoff?

Where to look in the EHR: Orders (Fall precautions, Out of bed to chair); Flowsheets > Safety and Mobility; PT note 1520; Pharmacy Transfer Medication Reconciliation

Select ALL that apply.

- ☐ Keep the bed low, the bed alarm on, the call light in reach and non-skid socks on.
- ☐ Assist x2 with a gait belt for any transfer to the chair or bathroom; the right arm stays elevated/splinted.
- ☐ Offer the urinal and a toileting schedule, since he will have an urge to void without the Foley and may get up alone.
- ☐ Let him walk to the bathroom alone now that the lines are out, because he is alert and oriented.

- ☐ Hold the oxycodone and gabapentin all day so that he is less sleepy.
- ☐ Turn the bed alarm off because he is oriented x4 and CAM-ICU negative.
- ☐ Teach him to call before getting up, and have him dangle at the edge of the bed before standing (he had mild dizziness when he first dangled at the edge of the bed on Wednesday).
- ☐ Apply a soft wrist restraint to keep him in bed.
- ☐ Hand off the fall risk and the PT/OT plan to the Med-Surg nurse.

6.12 Night I&O Totals *(0.75 pts)*

Open Intake/Output and read the line "Night shift 11/04 1900 to 11/05 0700". (a) Write the intake, output and net. Compute his average urine output in mL/h and in mL/kg/h (82 kg) over the 12 hours. (b) Compare this night with the night before (11/03 1900 to 11/04 0700) and say what the change means for his fluid status. (c) His weight is 89.8 kg this morning and 82 kg on admission: what does that tell you, and how will you track output now that the Foley is out?

Where to look in the EHR: Flowsheets > Intake/Output (shift totals); Flowsheets > Cardiac (daily weight)

Blood loss (escharotomy)

TOTALS

Totals	Intake mL	Output mL	Net mL
Day shift 11/03 0700 to 11/03 1900	4094	474	+3620
Night shift 11/03 1900 to 11/04 0700	4160	803	+3357
Day shift 11/04 0700 to 11/04 1900	1823	836	+987
Night shift 11/04 1900 to 11/05 0700	1257	638	+619
Day shift 11/05 0700 to 11/05 1900	102	100	+2
Last 24 hours (to now)	**2936**	**1506**	**+1430**
Whole stay (matches the Summary page)	**22253**	**4040**	**+18213**

Figure: the Totals section of Intake/Output as displayed in the EHR at 0700 Thursday.

6.13 The Final Handoff: Transfer SBAR to Med-Surg *(2.5 pts)*

It is 0650 on Thursday 11/05. Write your SBAR transfer handoff to C. Pruitt, RN (Med-Surg 2 East, burn step-down). Use real numbers and include: the burn infection signs to watch, central line day, the cefazolin schedule, what is still pending, the pain plan, mobility, Friday's OR and the NPO time, and the family contacts. Mention one safety item for the first hours after arrival.

Where to look in the EHR: Everything you read and charted this shift; Orders; MAR; Lab Results; Notes (Burn Surgery Transfer Note, ICU Transfer Summary, Case Management Note, Pharmacy Reconciliation)